

§13.2 Derivatives and Integrals of Vector Functions

Homework : 10,19,21,23,33

Definition : Let $r(t) = \langle f(t), g(t), h(t) \rangle$. Then

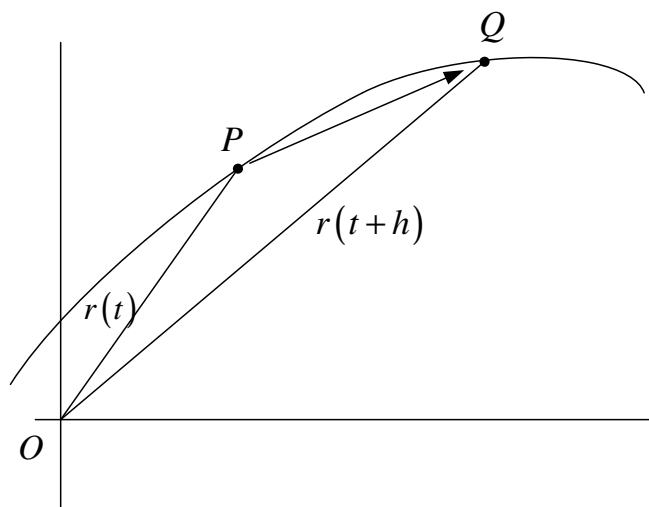
$$\Delta r(t) = \langle \Delta f(t), \Delta g(t), \Delta h(t) \rangle. \text{ Here}$$

$$\Delta = \lim_{t \rightarrow a}, \quad \frac{d}{dt} \quad \text{or} \quad \int$$

(極限) (微分) (積分)

(1) $r'(t)$ = the tangent vector to the curve defined by r at the point P .

(過 P 點的 r 曲線的切向量) = 在 t 時間的(瞬間)速度向量



$$\overline{PQ} = \overline{OP} - \overline{OQ} = r(t) - r(t+h) = \text{割向量}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{r(t) - r(t+h)}{h} = r'(t) = \text{過 } P \text{ 點的切向量.}$$

Example 1 : Let $x = t^5$, $y = t^4$, $z = t^3$, $t \in R$. Find the equation of tangent to the curve at $(1,1,1)$.

Solution :

$$r'(t) = \langle 5t^4, 4t^3, 3t^2 \rangle_{t=1} = \langle 5, 4, 3 \rangle.$$

Example 2 : Let $r(t) = \langle \cos t, \sin t, t \rangle$. Find

(i) velocity when $t = 2\pi$ (ii) distance traveled over $[0, 2\pi]$.

Solution :

$$(i) \quad v(t) = r'(t) = \langle -\sin t, \cos t, 1 \rangle_{t=2\pi} = \langle 0, 1, 1 \rangle$$

$$|v(t)| = \text{速度 (speed)} = \sqrt{2}$$

$$(ii) \quad d = \int_0^{2\pi} |v(t)| dt = 2\pi\sqrt{2} = 2\sqrt{2}\pi.$$

(2) Differentiation Rules

$$(i) \quad \frac{d}{dt}(u(t) \cdot v(t)) = u'(t) \cdot v(t) + u(t) \cdot v'(t)$$

$$(ii) \quad \frac{d}{dt}(u(t) \times v(t)) = u'(t) \times v(t) + u(t) \times v'(t)$$

$$(3) \quad \frac{d}{dt}|r(t)| = \frac{r'(t) \cdot r(t)}{|r(t)|}$$

$$\textbf{Proof :} \quad |r(t)|^2 = r(t) \cdot r(t)$$

$$\Rightarrow 2|r(t)| \frac{d}{dt}|r(t)| = 2r'(t) \cdot r(t)$$

$$\Rightarrow \frac{d}{dt}|r(t)| = \frac{r'(t) \cdot r(t)}{|r(t)|}$$